

# Practice Quiz: Agents -- You as a Prediction Machine

## Part A: Multiple Choice

1. In Active Inference, an **agent** is any system that: A) Follows instructions from an authority B) Maintains a generative model and acts to minimize free energy C) Has a physical body with legs and arms D) Maximizes reward at all costs

**Answer: B.** Agency is defined by having a boundary, a generative model, and the capacity to act on predictions.

1. Your sense of personal identity functions as a **generative model** because it: A) Is a fixed list of personality traits stored in your brain B) Generates predictions about who you are, what you value, and how you will behave C) Was assigned to you by your parents D) Never changes after childhood

**Answer: B.** Identity is an ongoing process of prediction and self-model maintenance, not a static list.

1. When you fail a math test despite believing "I am good at math," the resulting discomfort is best explained as: A) Bad luck B) Prediction error between your self-model and sensory evidence C) A sign you should give up math forever D) Proof that self-models are useless

**Answer: B.** The discrepancy between your self-prediction and reality creates free energy that must be resolved.

1. A student who chooses classes based on genuine curiosity rather than peer pressure has **higher agency** because: A) They are more popular B) Their actions are driven by their own generative model rather than external forces C) They always get better grades D) They ignore all social information

**Answer: B.** Agency is the degree to which your own internal model, rather than outside pressure, drives behavior.

1. Adolescence involves intense self-model revision, which feels uncomfortable because:  
A) Teenagers are inherently moody B) The old child self-model no longer fits, and the new adult model is not yet stable, causing high prediction error C) Schools make adolescence harder than it needs to be D) Hormones eliminate all generative models

**Answer: B.** Active Inference predicts that fundamental model revision is cognitively expensive and produces sustained prediction error.

1. You act differently around your parents, friends, and teachers. Active Inference explains this as: A) Being fake or hypocritical B) Having one generative model with context-dependent predictions for different social settings C) Evidence of multiple personalities D) A failure of your self-model

**Answer: B.** A sophisticated agent generates different appropriate predictions for different environments.

1. Cognitive dissonance -- feeling uncomfortable when your actions contradict your beliefs -- is an example of: A) Free energy being maximized B) Prediction error between your self-model and your observed behavior C) A Markov Blanket collapsing D) Random noise in the brain

**Answer: B.** When you act contrary to your self-predictions, the discrepancy generates discomfort that demands resolution.

1. Writing a college application essay is difficult because it requires you to: A) Memorize facts about the college B) Make your normally implicit self-model explicit, inspecting predictions that usually run below awareness C) Lie about your achievements D) Copy someone else's identity

**Answer: B.** Articulating your generative self-model forces conscious inspection of predictions that normally operate automatically.

1. Social media can be identity-destabilizing because: A) It wastes time B) Likes and comments dramatically increase the rate of prediction error feedback about your self-

model C) Algorithms delete your real personality D) It replaces your Markov Blanket with a digital one

**Answer: B.** The rapid feedback loop between your online persona and others' reactions accelerates self-model updating.

1. If identity is a generative model rather than a fixed essence, then personal growth means: A) Your atoms are being replaced B) Your generative self-model is being updated through accumulated prediction errors and revised predictions C) You are becoming a completely different person with no continuity D) Growth is impossible because models cannot change

**Answer: B.** Growth is the ongoing process of revising your self-model in response to new evidence about who you are.

## **Part B: Short Answer**

1. Describe a time when you discovered something about yourself that surprised you. What prediction did your self-model get wrong, and how did you update it?
2. How does the Active Inference view of identity explain why some people have an "identity crisis" during high school? What is happening to their generative model?
3. Give an example of resolving cognitive dissonance by dismissing evidence rather than updating your self-model. Why might this strategy backfire in the long run?
4. How much of your sense of identity comes from your own generative model versus the expectations others project onto you? Where do you draw the line?
5. If a friend says "you have changed," describe what is happening in Active Inference terms. Whose generative model is out of date -- yours or theirs?